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TWO COMPONENT MAGNETISATION IN IRISH CARBONIFEROUS LIMESTONE

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Many of the Lower Carboniferous Limestones of southern Ireland carry an appreciable remanent magnetisation. Samples of this rock have been subjected to careful A.C. and thermal demagnetisation. The former process isolates a stable component of magnetisation in the Permian field direction, whilst the latter reveals the presence of another component with a significantly different (Lower Carboniferous?) direction. A suitable distribution of near domain size magnetite grains is suggested as a possible source of these two components.

1. Introduction and method

Palaeomagnetic investigation of the Carboniferous limestones of southern Ireland [1, 2] reveals that in many regions rocks of this age carry a stable magnetisation in the direction of the Permian field. Permian remagnetisation is a feature of many European rocks of Palaeozoic age [3, 4] but frequently it is also possible to detect the presence of an older component, especially by the use of thermal demagnetisation. In

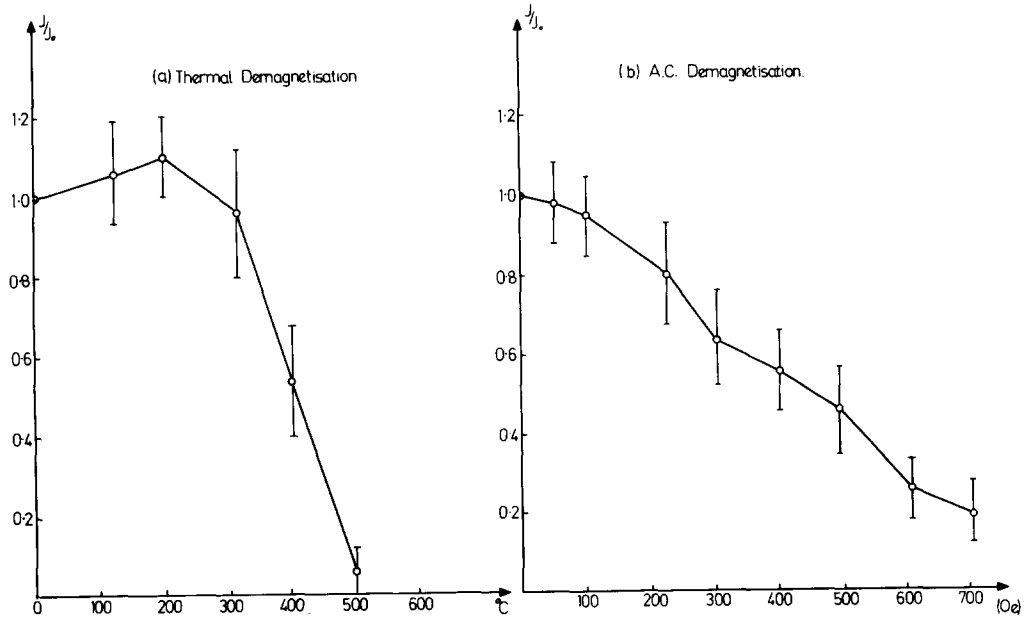


Fig. 1. Mean normalised demagnetisation curves for limestone samples from 20 sites.

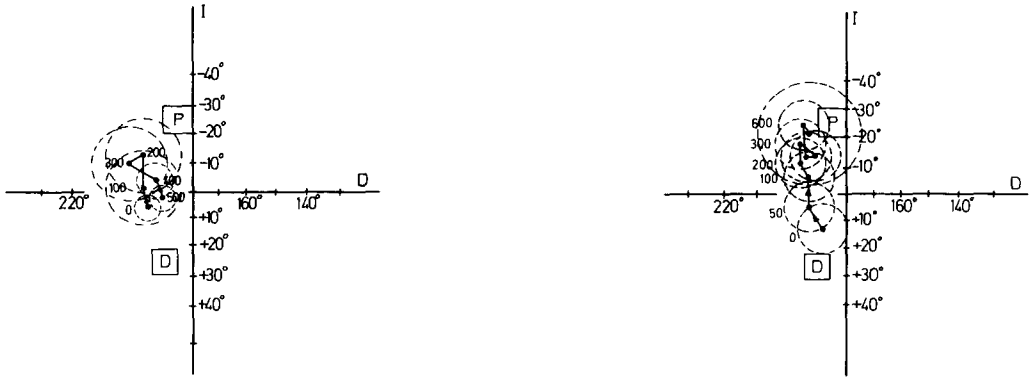


Fig. 2. Movement of mean direction of magnetisation of pilot specimens from 20 sites. (a) Thermal demagnetisation (°C). D and P are the approximate directions of the upper Devonian and Permian magnetic fields plotted from the data of Storetvedt (1968). (b) A.C. demagnetisation (Oe).

order to establish whether such a component existed within the Irish limestones of Lower Carboniferous age, laid down at a time during which the field direction was possibly quite different from that of the Permian [5], thermal and A.C. demagnetisation were carried out on pilot samples from some 30 sites. These sites – widely spaced over the southern part of Ireland to provide what could be considered a representative survey of the Z-age strata of the Lower Limestone – are described and mapped in ref. [1].

The thermal and A.C. demagnetisation curves obtained were all surprisingly similar and it is possible to construct with some confidence average curves for the two methods (fig. 1). Unlike red sediments, in which the magnetisation carried by haematite is generally very resistant to A.C. fields, the limestones are readily affected by fields of a few hundred oersted. The thermal curve shows a rapid decrease above 300°C, falling to a maximum blocking temperature at about 550°C.

The two methods reveal the change of direction of magnetisation within the samples in quite different ways (fig. 2). After the initial removal of a soft component in the present field direction, during which the magnetisation swings towards the Permian direction, the A.C. method swings the vector still further upwards whilst thermal treatment produces an abrupt change in trend and the vector swings downwards towards a positive inclination. As the precise nature of the thermal movement is somewhat in doubt owing to a wide scatter of data points which give 95% confidence circles of considerable radius, a more rigorous experiment was carried out to check the validity of the result.

A large orientated block sample, (92H) taken from a quarry at Goresbridge, Co. Kilkenny, was drilled to provide 34 specimens. The rock is a medium-grained, crinoidal, shelly, bioclastic limestone; abundant pyrite and a fair amount of ferroan dolomite are present but no magnetic minerals can be observed by the use of optical methods. Progressive demagnetisation in air was carried out using the furnace at the Close House laboratory of the University of Newcastle-upon-Tyne [6]. The earth's field was kept annulled as closely as possible, and the greatest variation from this state was an excursion of about 30γ. To ensure that any shift in the inclination of the magnetisation was not due to inaccurate cancellation of the earth's field during the demagnetisation process, half of the specimens were placed in an inverted position in the furnace. Both the upright and inverted specimens showed similar changes in direction of magnetisation at temperatures below 500°C so that these results would therefore appear to be genuine and not a function of the experimental method. At temperatures above 500°C the directions in the upright and inverted specimens moved in opposite ways, a result probably due to the impossibility of cancelling the field sufficiently to prevent the build-up of a small TRM in the extremely magnetic maghemite phase produced at these temperatures by the decomposition of the pyrite in the rock. For this reason, results obtained at temperatures of 500°C and above have been discarded.

The initial magnetisation of the samples from the block shows streaking-out towards the present field direction but, on heating to 250°C, the positions coalesced into a compact group centred close to the Permian field direction (fig. 3b). Seven of the speci-

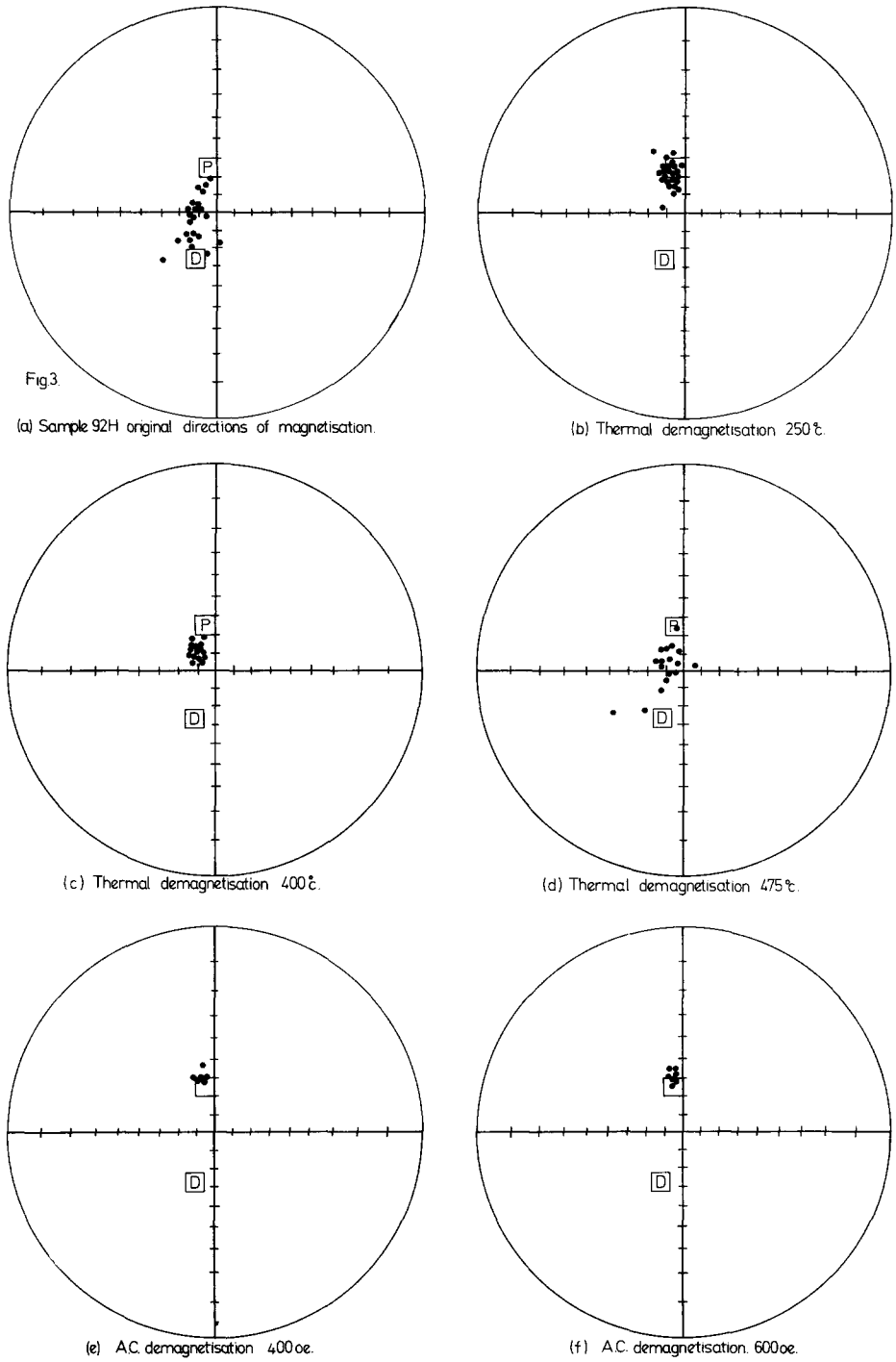


Fig. 3. (a) Sample 92H original directions of magnetisation. (b) Thermal demagnetisation 250°C. (c) Thermal demagnetisation 400°C. (d) Thermal demagnetisation 475°C. (e) A.C. demagnetisation 400 Oe. (f) A.C. demagnetisation 600 Oe.

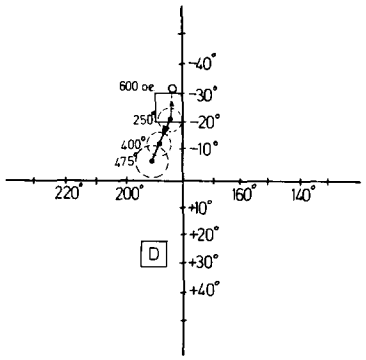


Fig. 4. Sample 92H. Movement of the mean direction during demagnetisation.

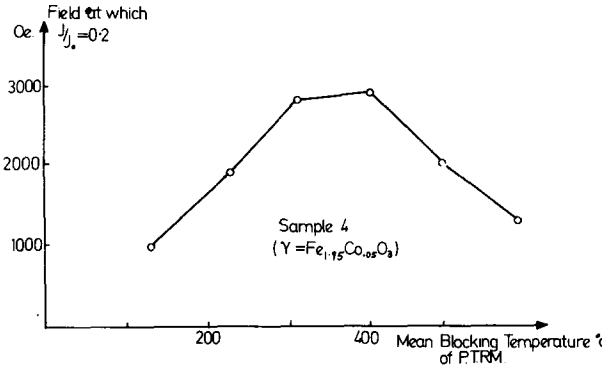


Fig. 5. Variation of coercivity with blocking temperature plotted from the data of Dunlop and West (1969).

mens were then A.C. demagnetised in fields of 400 and 600 Oe. A shift to a stable direction of 188° – 30° was produced (figs. 3e, 3f). Further heating of the remainder of the specimens resulted in the shift towards the positive inclination previously observed, though in this case the dispersion was much reduced. By 475°C many of the specimens were at the limit of resolution of the magnetometers, and the uncertainty in measurement probably accounts for some of the scatter displayed at this temperature. Fig. 4 shows the mean direction of magnetisation of the specimens during demagnetisation. Again a trend towards the positive during thermal treatment is clearly demonstrated.

In view of this evidence there seems little doubt that the limestone carries two stable components of magnetisation; the predominant component is a secondary magnetisation in the Permian field direction yet some portion of what is possibly a primary magnetisation of Lower Carboniferous age is also retained.

From the preceding results it will be obvious that, even though the A.C. method reduces the intensity of magnetisation of the sample to a level (say $J/J_0 = 0.2$) at which thermal treatment would have produced a distinct shift of the magnetisation towards a positive inclination, no such shift is in fact produced. It would seem that when using A.C. the 'Carboniferous' portion of the magnetisation is preferentially removed, whereas under thermal treatment it is the 'Permian' portion which is removed first, i.e. the portion of the magnetisation with high blocking temperatures has low coercive forces whilst the portion with low blocking temperatures has high coercive forces.

Discussion of the nature of the magnetic assemblage producing this effect is somewhat hindered by the fact that to date no magnetic mineral has been identified within the limestone. Any such mineral present must be in a very finely divided state, probably not much above domain size. A mixture of magnetite and haematite could be concocted to explain the results but there is no evidence to confirm the existence of haematite; the form of the demagnetisation curves suggests that magnetite is probably the dominant magnetic mineral. The question thus arises as to whether the effects described above could be explained by the presence of magnetite alone. Some evidence that grain distributions with the required properties do occur is to be found in the work of Everitt [7] and Dunlop and West [8]. The experiment which both these researchers carried out was to give samples partial TRMs over small discrete temperature ranges and then demagnetise these by the A.C. method. Both researchers found samples (the former a basalt from Kingshorn, Fifeshire, the latter a synthetic cobalt-doped maghemite) in which the coercive forces of the high temperature PTRMs were lower than those of lower temperature PTRMs. Fig. 5 is plotted from the data of Dunlop and West. If the coercivity-blocking temperature spectrum of the Irish limestone was in any way similar to this and the various components of the magnetisation were carried by the portions under the curve, as shown in fig. 6, it is obvious that A.C. demagnetisation to field H would leave only Permian components, whilst heat-

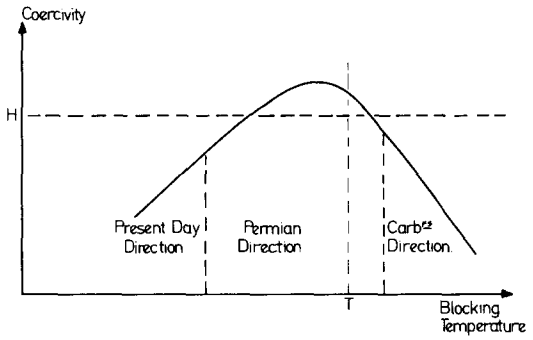


Fig. 6. Limestone samples: Suggested coercivity spectrum.

ing to temperature T would leave a predominant Carboniferous component. Unfortunately it was impossible to perform the PTRM experiment with the limestone, as the decomposition of the pyrite once again invalidated the results.

2. Conclusions

Lower Carboniferous limestones in Ireland carry two stable magnetic components, one in the Permian and one in the significantly different Lower Carboniferous field direction. The presence of the primary component can be shown by high temperature thermal demagnetisation whilst A.C. demagnetisation serves only to isolate the secondary component.

The magnetic minerals within the limestone are as yet unknown but the apparent phenomenon that the high coercivity portions of the magnetisation have lower blocking temperatures than the low coercivity portions could be explained either by the presence of a magnetite-haematite mixture or by a suitable distribution of magnetite grains alone.

Acknowledgements

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References

[1] P.Morris, Palaeomagnetic studies of the Carboniferous system in Ireland, Unpublished Ph.D. thesis, University of Dublin (1970).
[2] I.R.McAulay and P.Morris, Preliminary Palaeomagnetic results from some Irish Limestones of Carboniferous age, Sci. Proc. Royal Dublin Soc. Ser. A (1971) 45.
[3] F.H.Chamalaun, Origin of the secondary magnetisation of the Old Red Sandstone of the Anglo-Welsh cuvette, J. Geophys. Res. 69 (1964) 4327.
[4] J.Briden, Ancient secondary magnetisation in rocks, J. Geophys. Res. 70 (1965) 5205.
[5] A.Stephenson, Apparatus for thermal demagnetisation by the progressive method, in: Methods in Palaeomagnetism (Elsevier, 1967).
[6] K.Storetvedt, A synthesis of the palaeomagnetic data for Europe, Earth Planet. Sci. Letters 3 (1968) 444.
[7] C.Everitt, Thermoremanent magnetism II: experiments upon multidomain grains, Phil. Mag. 7 (1969) 583.
[8] D.Dunlop and G.West, An experimental evaluation of single domain theories, Rev. Geophys. 7 (1969) 709.